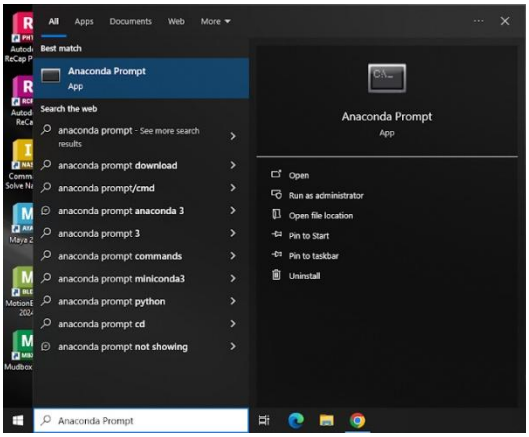
AERO60492 – Autonomous Mobile Robots

Coursework 2 – Path Planning – Installation Instructions

Section 1: Installation with Anaconda.

The following Instructions have been tested and found to work on cluster computers in Engineering Building A and George Begg, however Anaconda may not be installed on university computers in other buildings. **Do not use AppsAnywhere** for any of these instructions as this can cause issues. If you are using your own computer you can install Anaconda from here: https://docs.anaconda.com/free/anaconda/install/index.html.

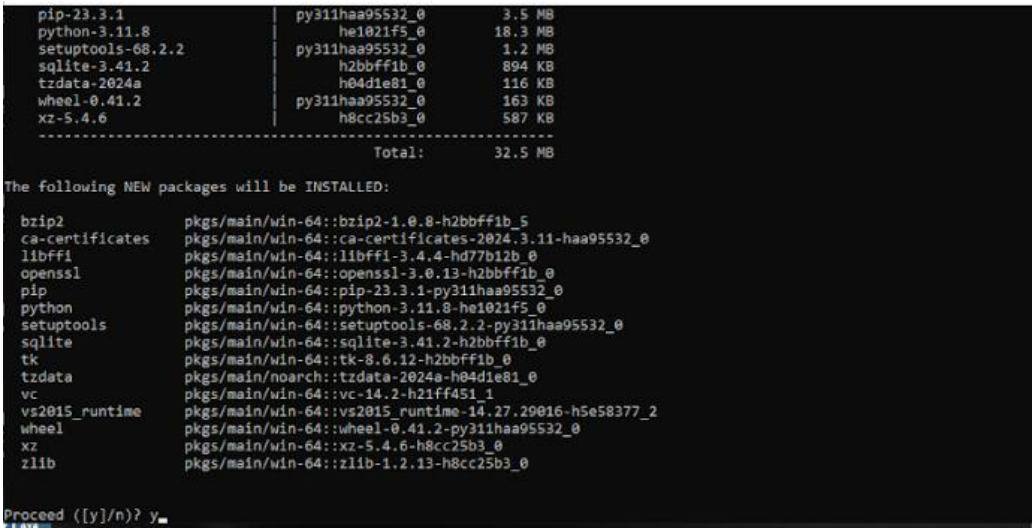
1. Download gui.py and pathPlanner.py from blackboard and put them in a folder on your P drive.
2. Launch Anaconda Prompt from the start menu.



3. Enter the following command to create a new environment, the python version is specified to ensure compatibility with the assignment program:

```
conda create -n AMR_assignment_2 python=3.11
```

When prompted, press y and the enter key to complete the environment creation:



```
  pip-23.3.1                 | py311haa95532_0        3.5 MB
  python-3.11.8              |      he1021f5_0       18.3 MB
  setuptools-68.2.2          | py311haa95532_0        1.2 MB
  sqlite-3.41.2              |      h2bbff1b_0        894 KB
  tzdata-2024a               |      h04d1e81_0        116 KB
  wheel-0.41.2               | py311haa95532_0        163 KB
  xz-5.4.6                   |      h8cc25b3_0        587 KB
  ------------------------------------------------------------
                                           Total:       32.5 MB

The following NEW packages will be INSTALLED:

  bzip2              pkgs/main/win-64::bzip2-1.0.8-h2bbff1b_5
  ca-certificates    pkgs/main/win-64::ca-certificates-2024.3.11-haa95532_0
  libffi             pkgs/main/win-64::libffi-3.4.4-hd77b12b_0
  openssl            pkgs/main/win-64::openssl-3.0.13-h2bbff1b_0
  pip                pkgs/main/win-64::pip-23.3.1-py311haa95532_0
  python             pkgs/main/win-64::python-3.11.8-he1021f5_0
  setuptools         pkgs/main/win-64::setuptools-68.2.2-py311haa95532_0
  sqlite             pkgs/main/win-64::sqlite-3.41.2-h2bbff1b_0
  tk                 pkgs/main/win-64::tk-8.6.12-h2bbff1b_0
  tzdata             pkgs/main/noarch::tzdata-2024a-h04d1e81_0
  vc                 pkgs/main/win-64::vc-14.2-h21ff451_1
  vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
  wheel              pkgs/main/win-64::wheel-0.41.2-py311haa95532_0
  xz                 pkgs/main/win-64::xz-5.4.6-h8cc25b3_0
  zlib               pkgs/main/win-64::zlib-1.2.13-h8cc25b3_0


Proceed ([y]/n)? y
```

4. Enter the following command to activate the environment you created in the previous step, note that every time you open anaconda prompt you will need to run this command to re activate the environment:

```
conda activate AMR_assignment_2
```

5. Next enter the following command to install PyQt5 which is required for the GUI to function:

```
pip install pyQt5
```

6. Now that the prerequisites are installed you will need to change directory to the location where you have saved the assignment.

```
cd C:<path to assignment_2 directory>
```

Where <path to assignment_2 directory> is where you have saved the assignment, for example \Users\username\Documents\AMR\AMR_assignment_2. If you are not sure of what the path is, find the directory in file explorer, right click and select 'Copy as path', you can then paste this path into Anaconda Prompt, make sure to remove the quote marks when doing this.

If you have not saved the assignment in the C drive, for example you are using the P drive, use the following command instead:

```
cd /d P:<path to assignment_2 directory>
```

8. To launch the GUI, run the following command:

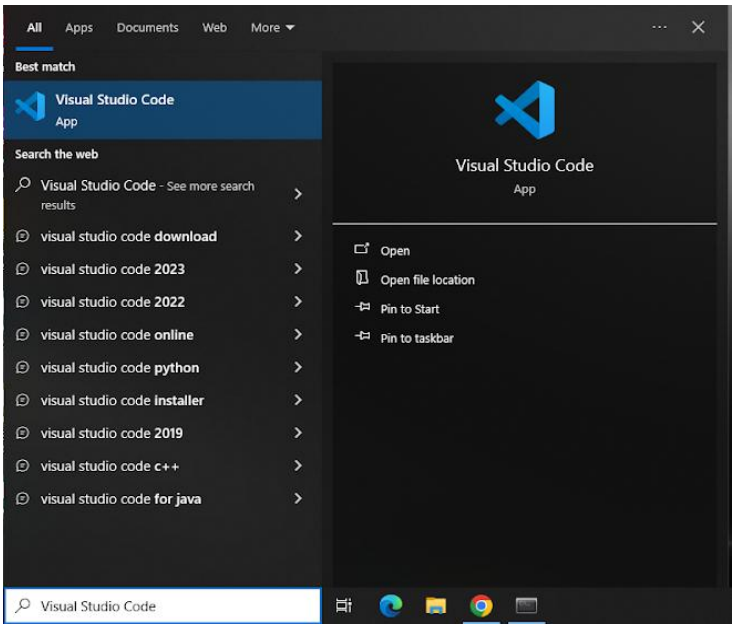
```
python gui.py
```

A user interface titled "AMR Coursework 2" should appear with a grid of squares and some buttons. The appearance and purpose of the buttons is given in the main Coursework 2 Document.

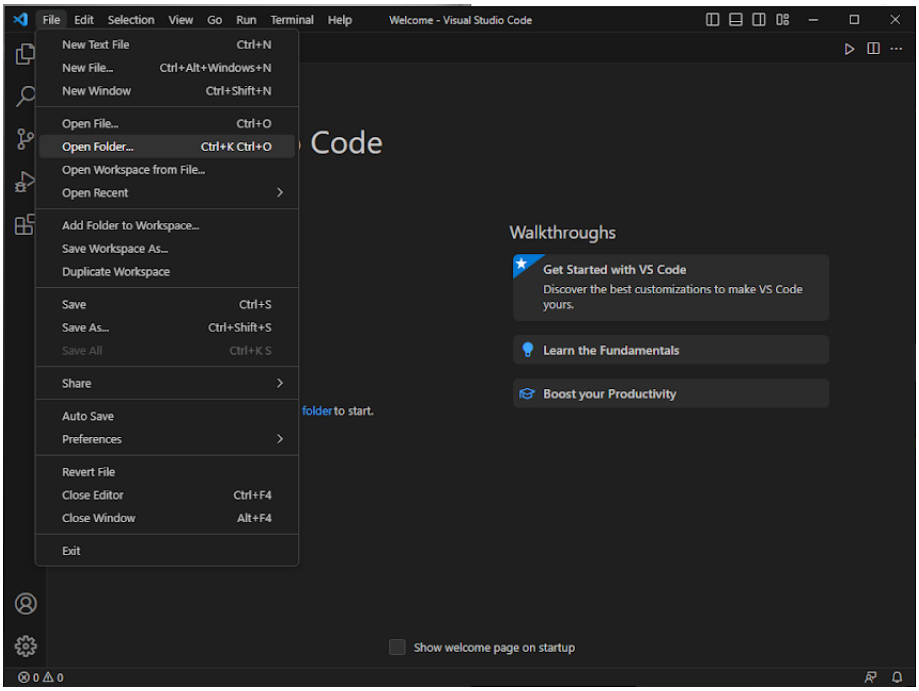
Section 2: Optional Instructions to set up Visual Studio Code

University cluster computers in Engineering Building A and George Begg will have Visual Studio Code installed, do not use AppsAnywhere to access VS Code as this can cause issues. If you are using your own computer Visual Studio code can be installed from https://code.visualstudio.com/download.

1. Launch Visual Studio code from the start menu:

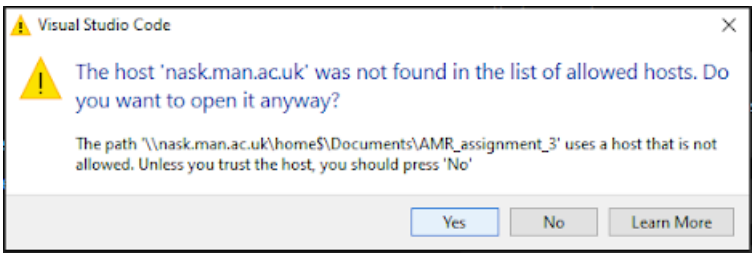


2. Click on **File** and then **Open folder**:

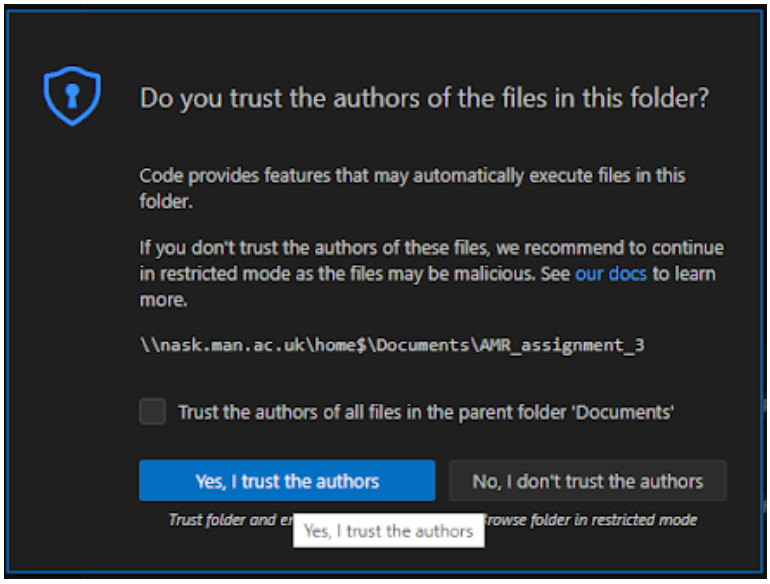


3. Navigate to the directory where you stored the .py files and click **Select Folder.**

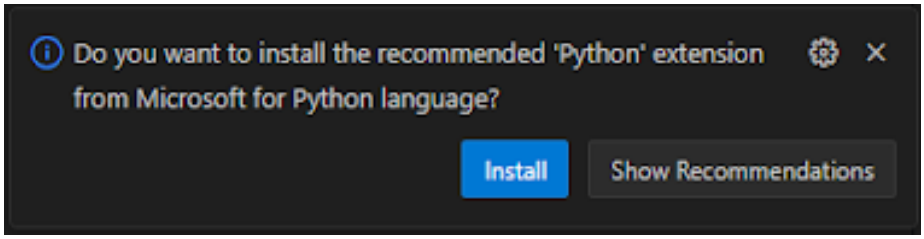
4. If you are greeted with the following dialog, select **Yes** to allow the folder to be opened:



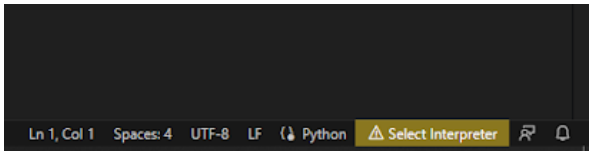
5. You will be asked if you trust the authors of this folder, select **Yes, I trust the authors**:



6. If you have not used Visual Studio Code for Python programming before you will be asked if you want to install the Python extension, click install:



7. Once the extension is installed, navigate back to the file explorer tab on the left of the screen and click on gui.py.
8. On the bottom right of the window, the right of "{} Python" it will either show the currently selected python interpreter e.g. "3.10.12 64-bit" or "Select Interpreter" in yellow:

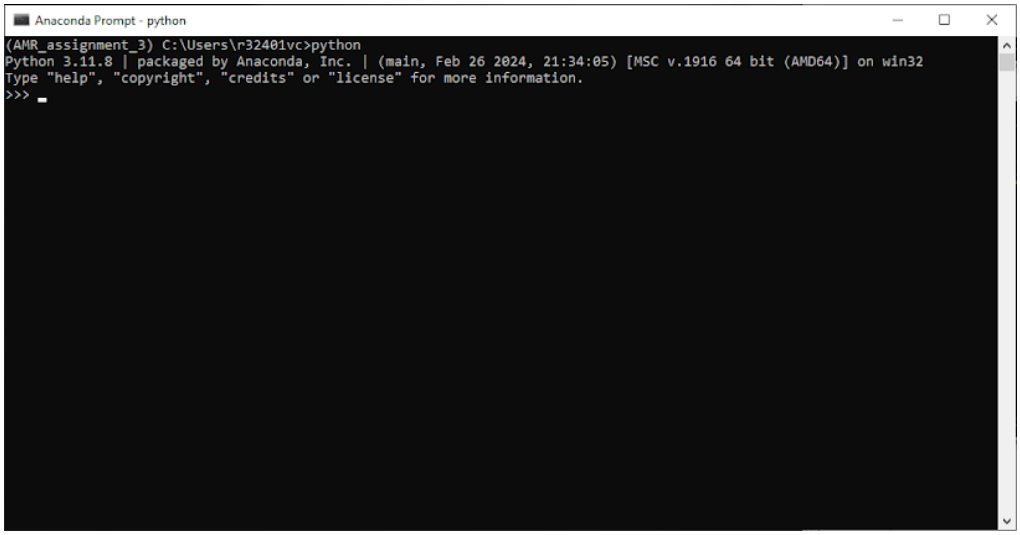


If an interpreter is shown, click on this and it will bring up a list of interpreters, select the one with the label "('AMR_assignment_2')", this is the conda environment you created in the installation instructions. You can now continue to step 11.

9. If "Select Interpreter" is shown in yellow then Visual Studio Code cannot find any python interpreters, we will need to locate the interpreter for the environment you created in step 2 of the installation instructions. Go back to Anaconda Prompt and make sure the environment is activated then enter the following command:

python

This will launch an interactive python session:



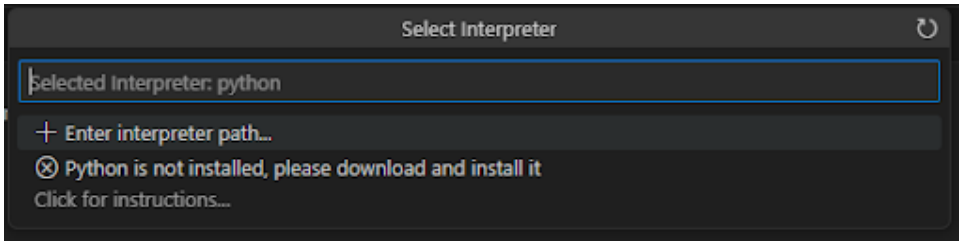
Type the following two commands into the python shell:

```
import sys
```

```
print(sys.executable)
```

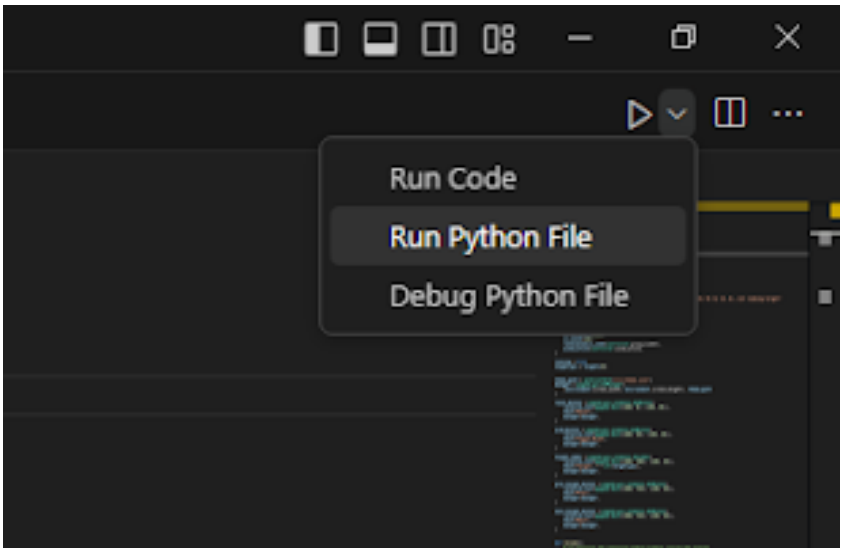
This will print out the location of the python interpreter, e.g. "C:\Anaconda3\envs\AMR_assignment_2\python.exe"

10. Go back to the Visual Studio Code window and click on "Select Interpreter", you will be presented with the following options:



Click on **+ Enter interpreter path…** and copy and paste the path you found in the previous step in the text box which appears and press enter.

11. When the file "gui.py" is selected, to run the program click the dropdown arrow next to the triangular play button in the top right corner and select "Run python file" to run the GUI:



Once you have completed the Visual Studio code setup instructions, you will only need to use step 11 to launch the simulation in future, Anaconda Prompt is no longer needed. Visual Studio Code will retain the same configuration every time it is opened.